

Maths2

7th guided works

Exercise 1 (Standard antiderivatives)

1. Determine all the antiderivatives (primitives) of the following functions on a well-chosen interval:

$$\begin{aligned} f_1(x) &= x^3 - 2x^2 + 6x - 7, & f_2(x) &= 2 \cos x - 3 \sin x, \\ f_3(x) &= x - e^x + 10, & f_4(x) &= \frac{5}{\sqrt{x}} - 2\sqrt{x} + \frac{3x^2 - 4x + 7}{x^3} \end{aligned}$$

2. If $F'_i(x) = f_i(x)$ for $i \in \{1, 2, 3, 4\}$, where f_i are the functions defined above. Find $F_i(x)$ for $i \in \{1, 2, 3, 4\}$ in the following cases :

$$F_1(0) = 1, \quad F_2\left(\frac{\pi}{4}\right) = 3\sqrt{2}, \quad F_3(0) = 1, \quad F_4(1) = 0$$

Exercise 2 (Integration by substitution method)

Using the substitution method, determine $F(x) = \int f(x) dx$ in the following cases :

$$\begin{aligned} (1) \quad f(x) &= \sin^7 x \cos x, \quad (2) \quad f(x) = x^2 \cos(x^3 + 1), \quad (3) \quad f(x) = \sqrt{2x + 1}, \\ (4) \quad f(x) &= e^{4x-3}, \quad (5) \quad f(x) = \frac{1}{(2x + 1)^3}, \quad (6) \quad f(x) = \frac{1 - \sin x}{x + \cos x} \end{aligned}$$

Exercise 3 (Integration by parts)

Using the integration by parts, determine $F(x) = \int f(x) dx$ in the following cases :

$$(1) \quad f(x) = e^{2x} \sin 3x, \quad (2) \quad f(x) = \arctan x, \quad (3) \quad f(x) = \arcsin x, \quad (4) \quad f(x) = \ln(2x + 1)$$

Exercise 4 (Integration of rational functions)

Integrate the following rational functions :

$$\begin{aligned} (1) \quad f(x) &= \frac{x}{9x^2 - 4}, \quad (2) \quad g(x) = \frac{2x - 3}{x^2 + 4}, \quad (3) \quad h(x) = \frac{x^2}{x^2 + 4}, \\ (4) \quad k(x) &= \frac{x - 1}{x^2(x^2 + 2x + 2)}, \quad (5) \quad l(x) = \frac{x^3}{x^2 - 3x + 2}, \quad (6) \end{aligned}$$

Exercise 5 (Definite integrals)

Calculate the following definite integrals :

$$\begin{aligned} (1) \quad \int_0^{\frac{\pi}{2}} (\sin^7 x \cos x) dx, \quad (2) \quad \int_{-1}^{\sqrt[3]{\frac{\pi}{4}-1}} x^2 \cos(x^3 + 1) dx, \quad (3) \quad h(x) = \int_0^1 \sqrt{2x + 1} dx, \\ (4) \quad \int_0^{\frac{3}{4}} e^{4x-3} dx, \quad (5) \quad \int_0^1 \frac{dx}{(2x + 1)^3}, \quad (6) \quad \int_0^\pi \frac{1 - \sin x}{x + \cos x} dx \end{aligned}$$

(You can use exercise 2!).